The Overlooked Robotics Supply Chain Is At An Inflection Point

By Bryce Smith

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As you know, we've had a large position in Tesla since 2019. It has stayed that way in large part because we recognize the massive potential for the Optimus Robot. In fact, our Ultimate Tesla Model shows that Optimus could generate over $55 billion in revenue by the year 2030, growing to over half a trillion in revenue by 2037. So we've long believed in the potential for humanoid robotics at Tesla to change the world.

But let's be clear, this Revolution is going to be much bigger than just Tesla. And more broadly speaking, The Robotics Revolution will be much bigger than just humanoids.

We all have to realize that this is such an insanely huge market -- perhaps the biggest ever -- and there are going to be many winners, including Tesla. There will be lots of different types of humanoids and perhaps hundreds of other form factors that will be huge in coming years, all enabled by the recent advances in AI. In fact, there is a good chance that the humanoid robotic market will probably look a lot like the auto market, with a handful of very successful original equipment manufacturers (OEMs) and many successful companies in the supply chain. There will certainly be millions (or billions!) of cheap, mass produced robots (like Optimus), but there will also be millions of high-end premium robots, specialty robots, working robots, luxury robots, etc. This is not going to be a winner take all market, and there is going to be lots of room for different companies to succeed.

Let's consider why we think The Robotics Revolution, and specifically The Humanoid Robotics Revolution, will be one of the biggest fundamental shifts in human history, and create an extraordinary amount of wealth in the process.

There are four key reasons why we think The Robotics Revolution has the potential to be the largest wealth creator in history:

1. Robots are the ultimate productivity engine. They perform physical tasks more efficiently, consistently, and safely than humans—and they never eat, sleep, or take vacations. By removing the dependence on human labor, robotics (similar to AI in white-color jobs) unlocks the single biggest constraint on economic growth: time worked.

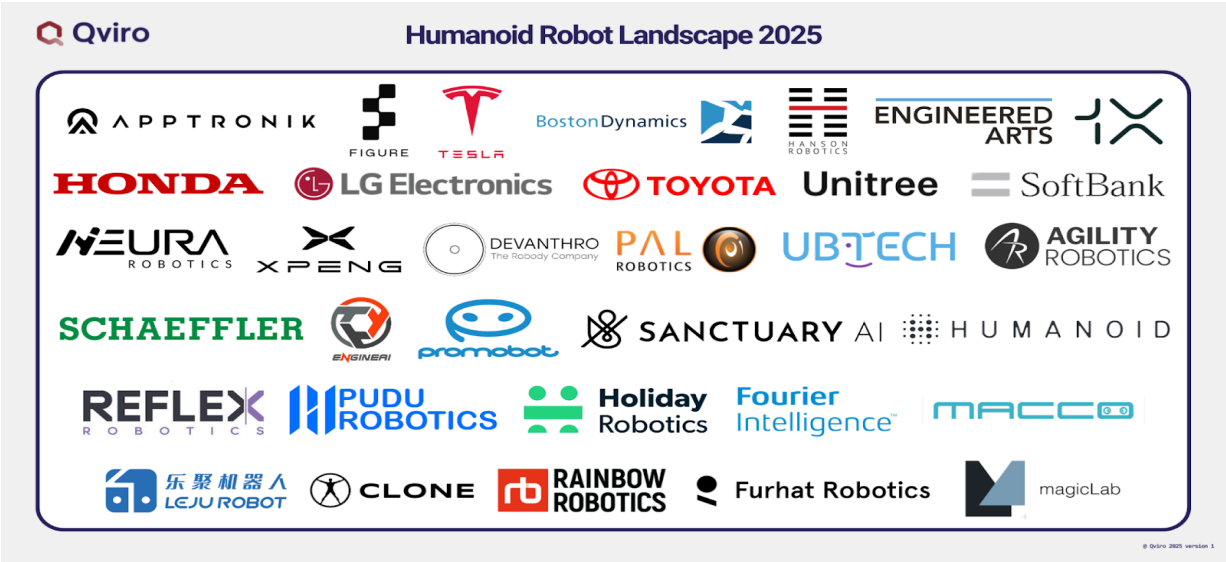
2. Beyond automating repetitive, boring, and dangerous work, robots will augment human capability—building lunar bases, assembling skyscrapers autonomously, exploring the deep ocean, and performing precision surgeries beyond human dexterity.

3. The humanoid form factor has a massive built-in advantage: the world is designed for human motion, tools, and environments. Humanoids can immediately leverage trillions of dollars of existing infrastructure—from door handles to forklifts—without requiring a redesign of the built world. Other form factors can be tailored to specific tasks where the humanoid is suboptimal or overkill.

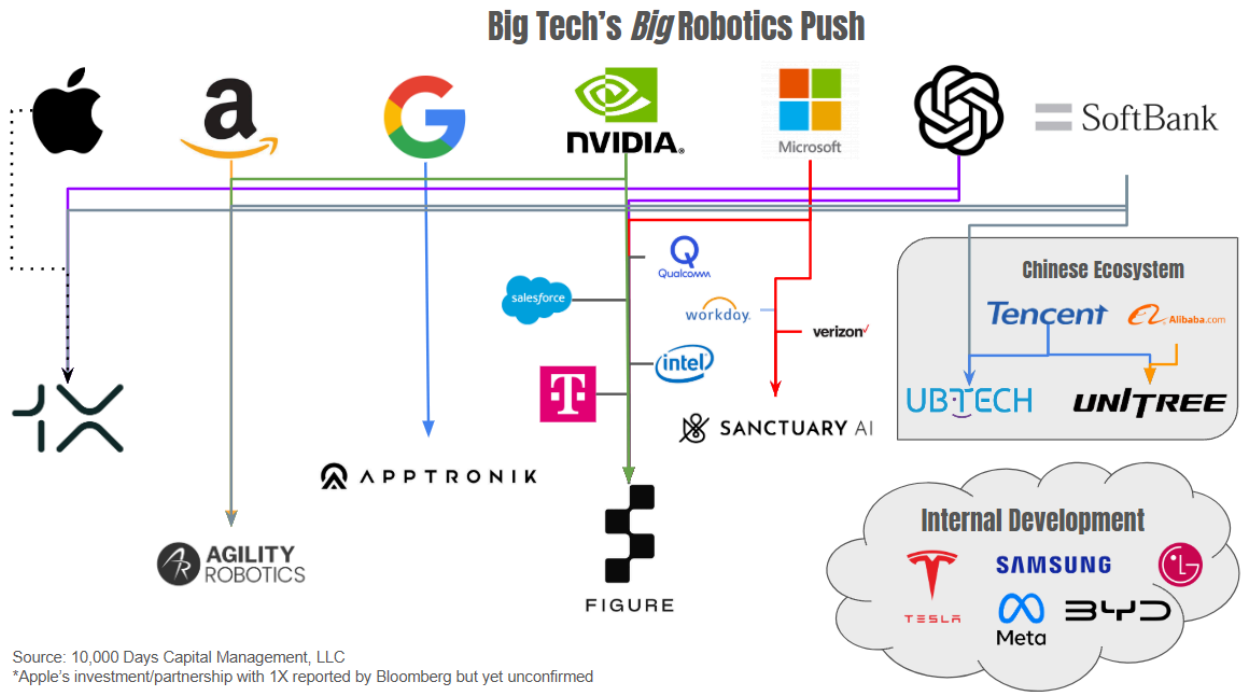
4. The total addressable market for robotics is effectively labor's share of global GDP—about $60 trillion in 2025. In the U.S. alone, $5–7 trillion of annual wages go to physical work. If robots capture even 10% of that, we're looking at a $500 billion annual market, minimum—comparable to the entire new car market today. Over time, that figure could scale into the multi-trillions as robot capabilities compound.

Critically, this Revolution is happening now because AI has made robots smart. For the first time, machines can learn tasks by observing them—just as humans do—instead of requiring every motion and decision to be hand-coded. This shift from programmed automation to learned behavior transforms robots from rigid tools into adaptable workers capable of continuous improvement.

So, how do we invest in this Revolution? First, we have and will continue to own an outsized position in Tesla. And while there are also some interesting humanoid robot startups, essentially all of these are privately held, so they are not an option for us as public-market investors. Nevertheless, it's impressive to see the growth in the number of major humanoid robotics startups in existence. There are now dozens of major humanoid builders, whereas just five years ago the number was basically one or two (essentially just Honda and Boston Dynamics).



Critically, many of these startups are working with major tech companies who are, in many cases, both investors and technological supporters. Below, we've assembled a diagram showing just some of big tech's investments in a select handful of humanoid robotics startups. **Cumulatively, these startups have raised north of $5 billion in funding and accumulated over $60 billion in market cap:**



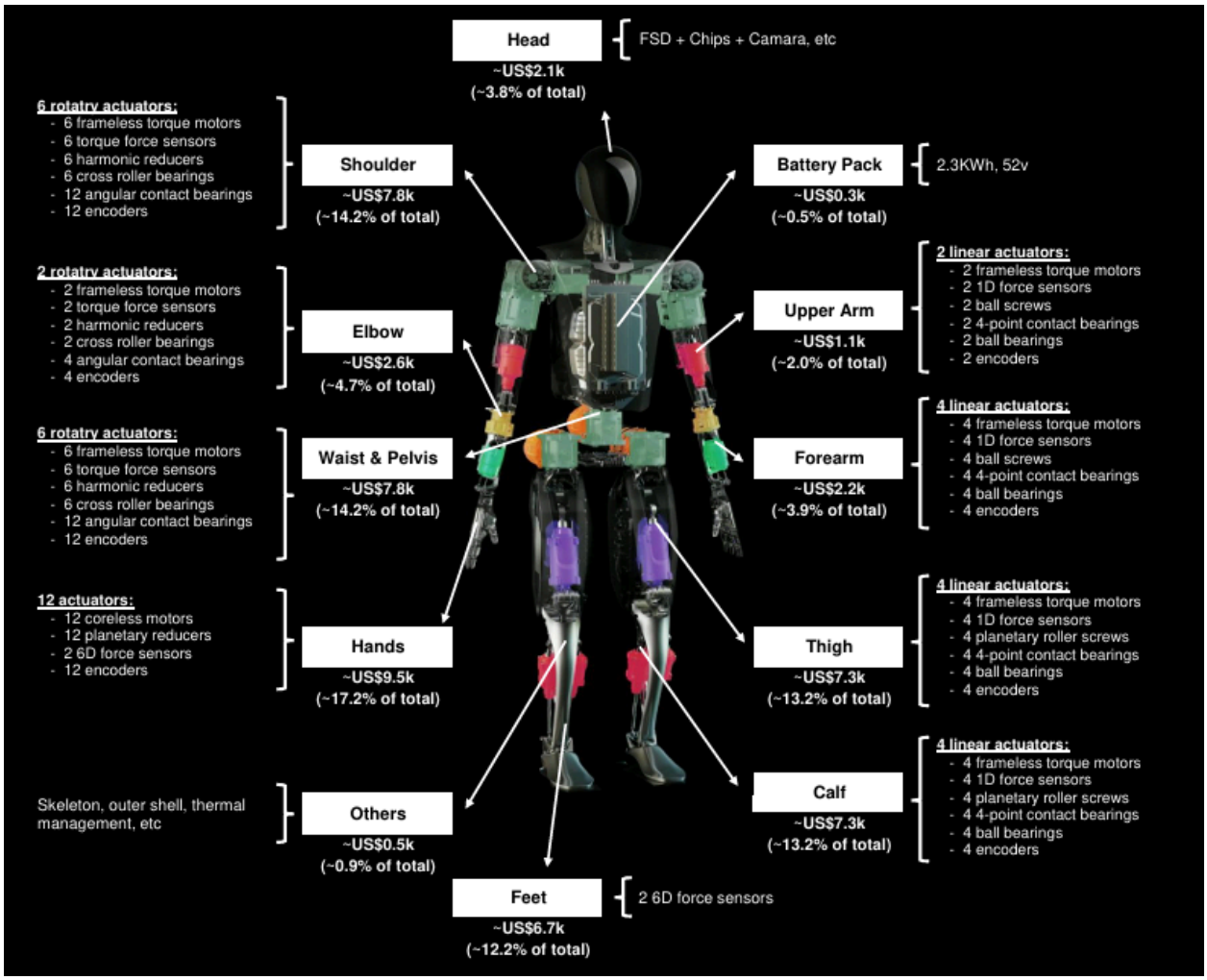
Source: 10,000 Days Capital Management, LLC
*Apple's investment/partnership with 1X reported by Bloomberg but yet unconfirmed

Big tech's investment in this space is critically important because it provides the financial runway for this nascent ecosystem to grow for the next few years. For big tech, robotics represents the next big paradigm shift following AI. As we've discussed, AI is enabling the rapid improvement in the functionality of robotics, and big tech sees this. Accordingly, while these companies compete to develop the best humanoid robots, most of them will be buying parts for their robots from our supply chain stocks.

Which leads us to our investment thesis: **the best non-Tesla way to invest in The Robotics Revolution at this moment is to invest in the robotics supply chain.** We're not exactly sure which of the companies above are going to successfully scale humanoid robots, but we are certain that basically all of these companies will buy parts from **the three public companies we've identified to be the best bets for this Revolution.**

So let's dig into what makes a humanoid robot, as we expect it will be the fastest growing segment of The Robotics Revolution over the next decade or so. Importantly, many of the parts that you need to build a humanoid robot are similar to what is used in other forms of robots. To

be clear, the humanoid supply chain is still nascent, and many of the companies that have been building parts for robots for years are only just now starting to develop humanoid-specific parts.



There are a lot of parts that go into a humanoid, from motors and actuators to sensors and batteries. I asked ChatGPT to estimate the total number of parts in a humanoid, and it came up with a range of 0.5-2 million parts, that's compared to about 30,000 parts for your typical automobile. Eventually, the number of parts will go down significantly as we get through the first few iterations of humanoid robots. The supply chain will start to consolidate and certain vendors will find a niche selling integrated systems that the OEMs need. Nevertheless, the sheer number of parts that are needed compared to autos gives me confidence that even if the OEMs vertically integrate, it will be very hard for them to produce all of these distinct parts internally.

In sum, Morgan Stanley estimates that the total BOM (bill of materials) for a humanoid robot (excluding the cost of software) to be in the range of $50-$60k per unit. I actually think that's too conservative, as I've seen estimates for the BOM as high as $300,000 for early prototypes.

Nevertheless, it's clear that there is a big opportunity for the supply chain companies if they can get their parts designed into these Revolutionary new robots.

As you can tell, we are very excited about the potential for the robotics supply chain. Almost no one we read or listen to in the mainstream financial media is talking about the massive potential for The Robotics Revolution, beyond the high-profile financing rounds for companies like Figure. We like this idea because it is platform agnostic, i.e. we don't have to pick the winning robot (even though we think that Tesla's Optimus is probably the most likely to go mainstream).

Further, it's also worth mentioning that we've followed this playbook before with The Internet Revolution, The Smartphone Revolution, and The App Revolution, among others. For example, we made great trades in some of the suppliers to each of those Revolution because we saw the coming respective inflection points. We think the same thing is about to happen with The Robotics Revolution.

We analyzed about fifty companies that might supply materials to the humanoid robotics (and other form factors) builders. What's exciting about these companies is that most of them are old-school industrial companies and they have a very long track record of profitable, stable growth. However, our job is not just to buy any company that fits the theme we are interested in, so we screened for the very best suppliers we could find at this moment. We identified three key suppliers to The Humanoid Robotics Revolution that have the potential to 10x, 50x, or even 100x from here.

Of course, there are risks involved. The two primary risks here are OEM vertical integration and cheap Asian competition, but we think current valuations of the stocks we've identified more than offset both. Tesla is the one player with the resources, culture, and obsession to vertically integrate humanoids end-to-end, so we're deliberately targeting suppliers with genuinely differentiated tech that Tesla can't easily replace. Moreover, most other humanoid OEMs simply don't have Tesla-level engineering depth or capital—Apptronik outsourcing its entire build to Jabil is the tell—and humanoids are far too complex for broad, full-stack verticalization anyway. As for China, yes, low-cost manufacturers can copy many components, but we expect a structural premium for Western-made parts because humanoid robots touch national security, enterprise IP, and eventually people's homes. Western OEMs aren't going to stuff their flagship robots with critical components sourced from non-allied countries.

In conclusion, we think the robotics supply-chain stocks we've are all trading at reasonable valuations and are about to see a big inflection in growth rates that the market isn't pricing in at all. Almost no one we read or listen to in the mainstream financial media is talking about the massive potential for The Robotics Revolution, beyond the high-profile financing rounds for companies like Figure.